Seat No.: _____ Enrolment No._____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–IV(NEW) EXAMINATION – WINTER 2022

Subject Code:3144202 **Date:16-12-2022**

Subject Name:Discrete Mathematics and Graph Theory

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

			MARKS
Q.1	**(a)**	Consider the complete graph K_n (1) Find the number m of edges in K_n. (2) Find the degree of each vertex in K_n. (3) Find those values of n for which K_n is regular.	**03**
	(b)	Let A be a set of factors of positive integer m and relation is divisibility on A. For m = 30, show that POSET $(A, \leq)$ is a Lattice	**04**
	(c)	(1) For $A = \{2,3,4,5,6\}$, $B = \{3,4,5,6,7\}$, $C = \{4,5,6,7,8\}$ find (a)$(A \cup B) \cap (A \cup C)$ (b)$(A \cap B) \cup (A \cap C)$ (2) If A= {2,3} B= {3,4} C= {2,4} find (a) (A X B) U (A X C) (b) (A X B)∩ (A X C)	**07**
Q.2	**(a)**	Let D_m denote the positive divisor of m ordered by divisibility. Draw the Hasse diagrams of D_{60}	**03**
	(b)	Let Z be the set of integers and an integer $m > 1$. Relation is defined as $a \equiv b(mod\ m)$ if $m \mid a - b$ Show that this relation defines equivalence relation on Z	**04**
	(c)	Solve the recurrence relation $a_{r+2} + 2a_{r+1} - 3a_r = 4$ using the method of undetermined coefficients.	**07**
		OR	
	(c)	Solve the recurrence relation $a_n = 5a_{n-1} - 6a_{n-2} + 7^n$	**07**
Q.3	**(a)**	Rewrite the following statements using quantifier variables and predicate symbols: (a) Some men are genius (b) Some numbers are not rational (c) Each integer is either even or odd	**03**
	(b)	Let $A = \{a, b, c, d\}, B = \{1,2,3,4\}$ & $C = \{u, v, w\}$ Define function $f: A \to B$ & g: $B \to C$ by $f = \{(a, 2), (b, 1), (c, 3), (d, 4)\}$, $g = \{(1, v), (2, u), (3, w), (4, v)\}$ Find the composite function gof.	**04**
	(c)	If p, q and r are three statements, using truth table show that i) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ ii) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$	**07**
		OR	
Q.3	**(a)**	Determine the validity of argument given: S_1: If I like Maths then I will study S_2: Either I will study or I will fail ---------- S: If I fail then I do not like Maths.	**03**
	(b)	Find DNF of the form $(p \to q) \wedge (q \to p)$without using truth table.	**04**
	(c)	Show that $p \leftrightarrow q$ is logically equivalent to$((p \to q) \wedge (q \to p))$ also $(p \to q) \wedge (q \to p)$ is logically equivalent to $(\sim p \vee q) \wedge (\sim q \vee p)$	**07**

Q.4	**(a)**	Define: Ring, Integral Domain, Field	**03**
	(b)	Define Subgroup of a group Check If $H = \{1, -1\}$ and $G = \{1, -1, i, -i\}$ then $(H,*)$ is a sub group $(G,*)$ or not?	**04**
	(c)	Let $R = \{0°, 60°, 120°, 180°, 240°, 300°\}$ and $* =$ binary operation, so that for and b in R $a * b$ is overall angular rotation corresponding to successive rotations by a and then by b. Show that $(R,*)$ is a group.	**07**

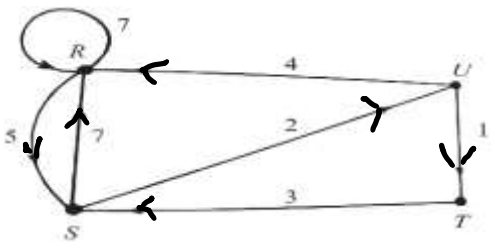
OR

Q.4	**(a)**	Show that $(Z_6, +_6)$ is cyclic group generated by $\bar{1}$ *and* $\bar{5}$.	**03**
	(b)	Let the function $f: R \to R$ be given by $f(x) = 2x - 3$. f^{-1} exists? If yes then Find f^{-1}	**04**
	(c)	State and prove Lagrange's theorem.	**07**

Q.5	**(a)**	Draw the non-directed multigraph corresponding to each of the following adjacency matrices	**03**

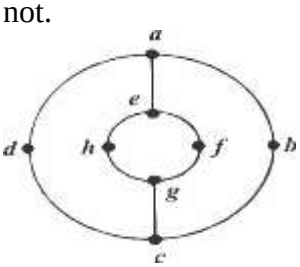
$$\begin{bmatrix} 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & 2 \end{bmatrix}$$

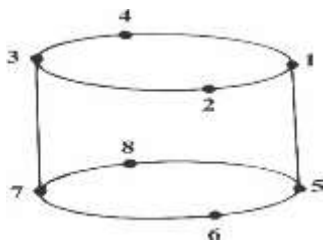
	(b)	Check the graph $K_{3,3}$ is planar or not ?	**04**
	(c)	Find the shortest distance matrix between each pair of vertices for a given digraph using Warshall's Algorithm.	**07**



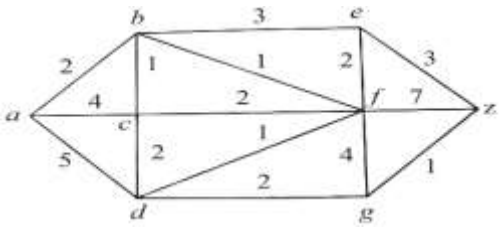
OR

Q.5	**(a)**	Is it possible to construct a graph with 12 nodes such that 2 of the nodes are having degree 3 and remaining vertices are having degree 4.	**03**
	(b)	Define isomorphic graphs. State whether the following graphs are isomorphic or not.	**04**





	(c)	Using Dijkstra's algorithm, find the shortest path between vertices a and z	**07**



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